

DL-Ops Assignment–1

Roll Number: **M25CSA012**

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1 Introduction

In this assignment, I observed variations in model architectures (ResNet18, ResNet50), optimisers (SGD, Adam), learning rates, batch sizes, and computational resources (CPU vs. GPU). Performance is evaluated using classification accuracy, training time, and FLOPs. I used the MNIST and FashionMNIST datasets for these experiments.

2 Datasets and Experimental Setup

2.1 Datasets

- **MNIST**: Handwritten digit dataset with 10 classes.
- **FashionMNIST**: Clothing image dataset with 10 classes.

Both datasets are split into **70% training, 10% validation, and 20% testing**.

2.2 Models Used

- ResNet-18 (pretrained = False)
- ResNet-50 (pretrained = False)

2.3 Training Configuration

- Optimizers: SGD, Adam
- Batch Sizes: 16, 32
- Learning Rates: 0.001, 0.0001

3 Q1(a): Deep Learning Model Performance

3.1 Test Classification Accuracy

Table 1: Test Classification Accuracy (%) on MNIST

Batch Size	Optimizer	Learning Rate	ResNet-18	ResNet-50
16	SGD	0.001	99.22%	99.14%
16	SGD	0.0001	97.46%	98.90%
16	Adam	0.001	98.79%	98.84%
16	Adam	0.0001	98.24%	98.89%
32	SGD	0.001	98.91%	98.84%
32	Adam	0.0001	96.91%	98.89%

Table 2: Test Classification Accuracy (%) on FashionMNIST

Batch Size	Optimizer	Learning Rate	ResNet-18	ResNet-50
16	SGD	0.001	91.18%	89.11%
16	SGD	0.0001	91.14%	91.10%
16	Adam	0.001	93.24%	88.84%
16	Adam	0.0001	92.19%	91.89%
32	SGD	0.001	92.17%	89.14%
32	Adam	0.0001	91.13%	92.07%

3.2 Analysis

Best test accuracy i got for MNIST dataset is 99.22% by ResNet-18(SGD, lr=0.001) and for FashionMNIST dataset is 93% by ResNet-50(ADam, lr=0.001). In this part of Question Adam helped the model learn faster in the early stages, especially on the harder FashionMNIST dataset, while SGD gave more stable and slightly better results on the simpler MNIST task. Increasing the depth to ResNet-50 did not help and instead led to overfitting, showing that a smaller, well-tuned ResNet-18 was a better fit for this problem.

4 Q1(b): SVM Classifier Performance

Table 3: SVM Classification Accuracy and Training Time

Kernel	Dataset	Accuracy (%)	Training Time (ms)
RBF	MNIST	96.20%	6031.92
Poly	MNIST	95.45%	6129.23
RBF	FashionMNIST	86.50%	5623.30
Poly	FashionMNIST	82.95%	6101.91

4.1 Analysis

The RBF kernel outperforms the polynomial kernel in most cases due to its ability to model non-linear decision boundaries. However, this comes at the cost of increased training time.

5 Q2: CPU vs GPU Performance Analysis

Table 4: Performance Comparison on FashionMNIST (CPU vs GPU)

Compute	Batch Size	Optimizer	LR	Accuracy (%)	
				ResNet-18	ResNet-50
CPU	16	SGD	0.001	85.02%	82.54%
CPU	16	Adam	0.001	85.58%	82.27%
GPU	16	SGD	0.001	87.36%	82.88%
GPU	16	Adam	0.001	86.99%	80.90%

Table 5: Training Time (ms) on FashionMNIST

Compute	Batch Size	Optimizer	LR	Training Time (ms)	
				ResNet-18	ResNet-50
CPU	16	SGD	0.001	327273.0	887577.1
CPU	16	Adam	0.001	532474.4	1204471.9
GPU	16	SGD	0.001	56131.5	104930.4
GPU	16	Adam	0.001	51225.7	110957.5

Table 6: FLOPs on FashionMNIST

Compute	Batch Size	Optimizer	LR	Accuracy (%)	
				ResNet-18	ResNet-50
CPU	16	SGD	0.001	1.83e+09	4.14e+09
CPU	16	Adam	0.001	1.83e+09	4.14e+09
GPU	16	SGD	0.001	1.83e+09	4.14e+09
GPU	16	Adam	0.001	1.83e+09	4.14e+09

6 Graphs and Visual Analysis

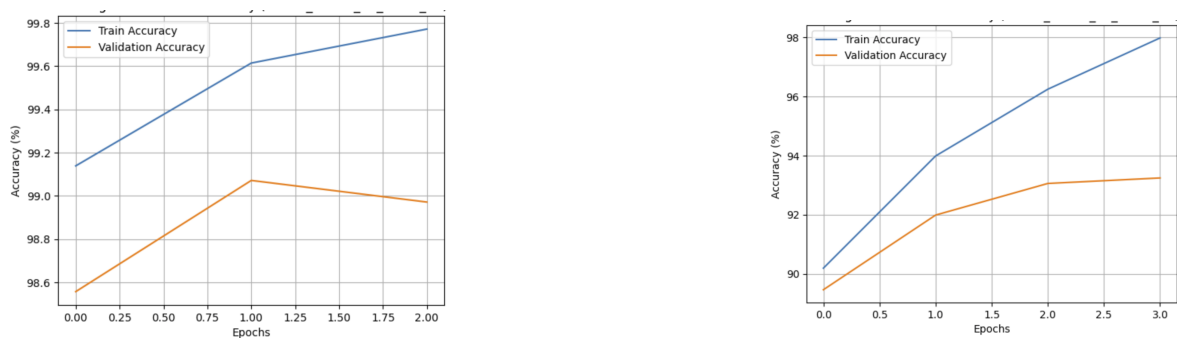


Figure 1: Best model accuracy graph for MNIST & FashionMNIST

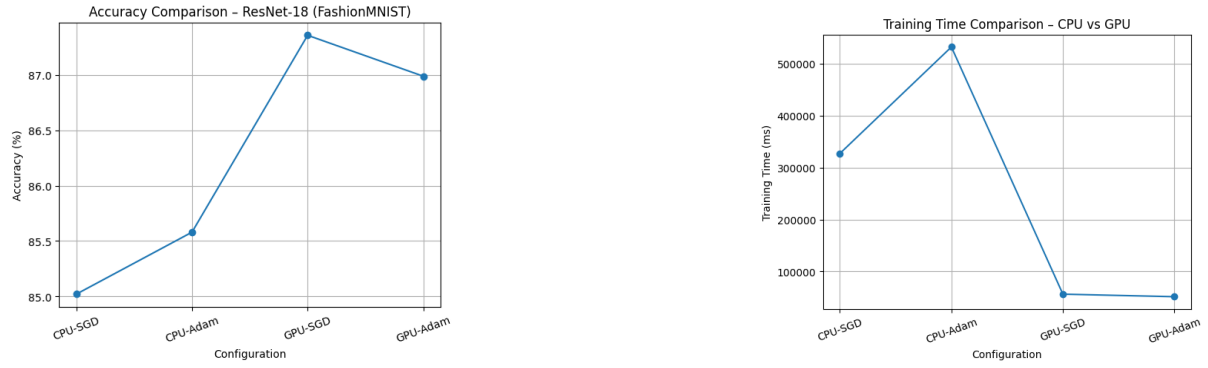


Figure 2: Accuracy and Time comparison between CPU and GPU

7 Conclusion

The experiments demonstrate that GPU-based training significantly reduces training time while improving accuracy. Adam optimizer provides faster convergence, whereas SGD offers stable performance. and also observed comparison between CPU and GPU. GPU is way faster then CPU for large computation.

GitHub and Colab Link

- GitHub Repository: https://github.com/jahanvi0106/MLOps-Jahanvi_Gajera-M25CSA012
- Colab Notebook: <https://colab.research.google.com/drive/15CBtp3gPoqMVlrm5LhQDBEndz02S7yMQ?usp=sharing>
- GitHub webpage: https://jahanvi0106.github.io/MLOps-Jahanvi_Gajera-M25CSA012/